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Regression Analysis

Business Analytics

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Chapter 6 Learning Objectives (LOs)

LO 1 Estimate and interpret a linear regression model.

- R^2 and Adjusted R^2

LO 2 Interpret goodness-of-fit measures.

LO 3 Model selection

LO 4 Test of Significant

- F-test

The Linear Regression Model

One of the most widely used techniques in **predictive analytics**.

- Captures the relationship between two or more variables.
- Predict an outcome of a target variable based on several input variables
 - Basically, detect **correlation**, not causality.
- Make assessments and robust predictions by determining which of the relationships matter the most and which we can ignore

Introductory Case: College Scorecard (1/2)

- College costs and student debt are on the rise
- Students and parents struggle to find clear, reliable data on critical questions of college affordability and value.
- The Department of Education (DOE) published a redesigned College Scorecard that reports the most reliable national data on college costs and students' outcomes at specific colleges.
- Fiona Schmidt, a college counselor, believes that the information from the College Scorecard can help her as she advises families.
- Fiona wonders what college factors influence post-college earnings and wants answers to the following questions:
 - If a college costs more or has a higher graduation rate, should a student expect to earn more after graduation?
 - If a greater percentage of the students are paying down debt after college, does this somehow influence post-college earnings?
 - And finally, does the location of a college affect post-college earnings?

Introductory Case: College Scorecard

- Information from 116 colleges on the following variables:
 - Annual post-college earnings (Earnings in \$)
 - The average annual cost (Cost in \$)
 - The graduation rate (Grad in %)
 - The percentage of students paying down debt (Debt in %)
 - Whether or not a college is located in a city (City equals 1 if a city location, 0 otherwise)
- Use the information to:
 - Make predictions for post-college earnings using regression analysis
 - Interpret goodness-of-fit measures for the post-college earnings model
 - Determine which factors are statistically significant in explaining post-college earnings

School ^a	Earnings	Cost	Grad	Debt	City
St. Ambrose Univ.	44800	22920	62	88	1
Albion College	45100	23429	73	92	0
⋮	⋮	⋮	⋮	⋮	⋮
Wittenburg Univ.	42700	26616	64	90	1

Notes: All data were downloaded from <https://collegescorecard.ed.gov> from May 24–26, 2016; see website for complete description of data.

The Linear Regression Model

- Develop a mathematical model that captures the relationship between the response variable y and the k predictor variables $x_1, x_2, \dots, x_k$.
- The model must also account for the randomness that is a part of real life.
 - Start with a **deterministic component** that approximates the *relationship*
 - Then consider a random *error* term to it, if making **stochastic model**.
- Use economic theory, intuition, and statistical measures to determine which predictor variables might best explain the response variable.

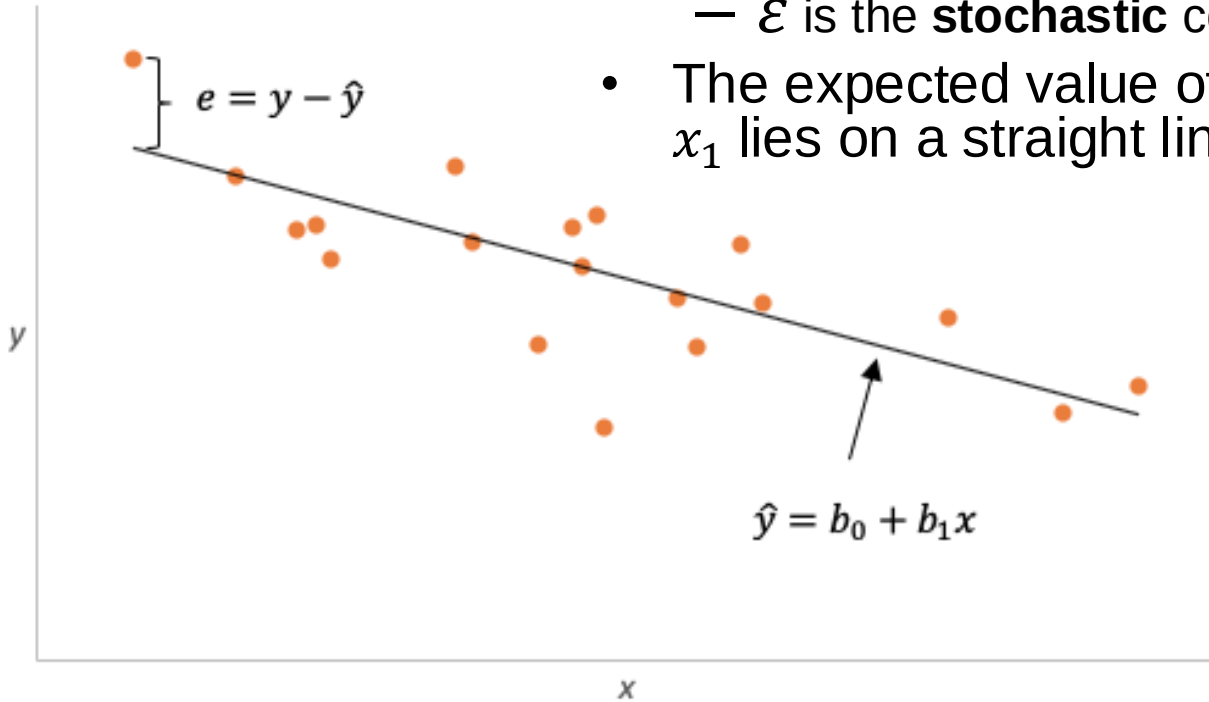
The Linear Regression Model

- A **simple linear regression** model uses only one predictor variable.

$$y = \beta_0 + \beta_1 x_1 + \varepsilon$$

- β_0 is the unknown intercept
- β_1 is the unknown slope
- $\beta_0 + \beta_1 x_1$ is the **deterministic** component
- ε is the **stochastic** component or random error term

- The expected value of y for a given value of x_1 lies on a straight line: $E(y) = \beta_0 + \beta_1 x_1$.

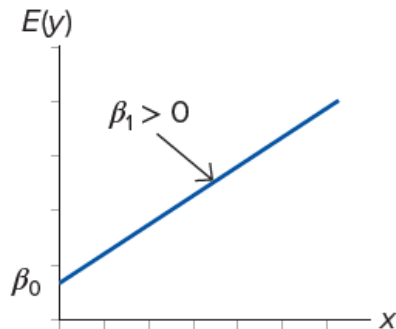


Simple Linear Regression: Manual computation

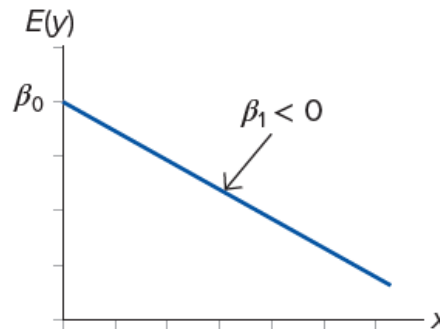
- Based on Ordinary Least Square (OLS) model, minimize the Sum Square of Error (SSE), using partial derivative:
 - For the simple case of one input, estimated $\hat{y} = B_0 + B_1x$
- Derivative for b_0 : $SSE = (y - \hat{y})^2 = \sum (y - B_0 - B_1x)^2$
 $d(SSE) / d(B_0) = -2\sum(y - B_0 - B_1x) = 0 \rightarrow B_0 = (\sum y - B_1\sum x) / n$
- Derivative for b_1 : $d(SSE) / d(B_1) = -2\sum x (y - B_0 - B_1x) = 0$
 $B_1 = (n\sum xy - \sum x \sum y) / (n\sum x^2 - (\sum x)^2)$
- Example: $X = [5, 15, 25, 35, 45, 55]$
- $Y = [5, 20, 14, 32, 22, 38]$
- Obtain $B_1 = 0.54$, then $B_0 = 5.63$ (See the Excel for detail computation)

The Linear Regression Model

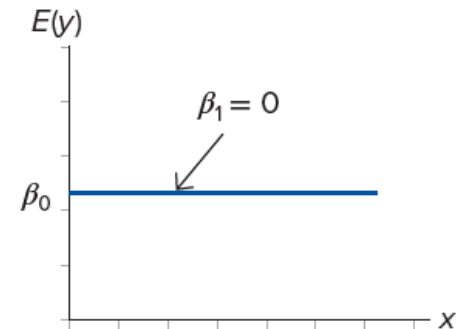
Positive linear relationship



Negative linear relationship

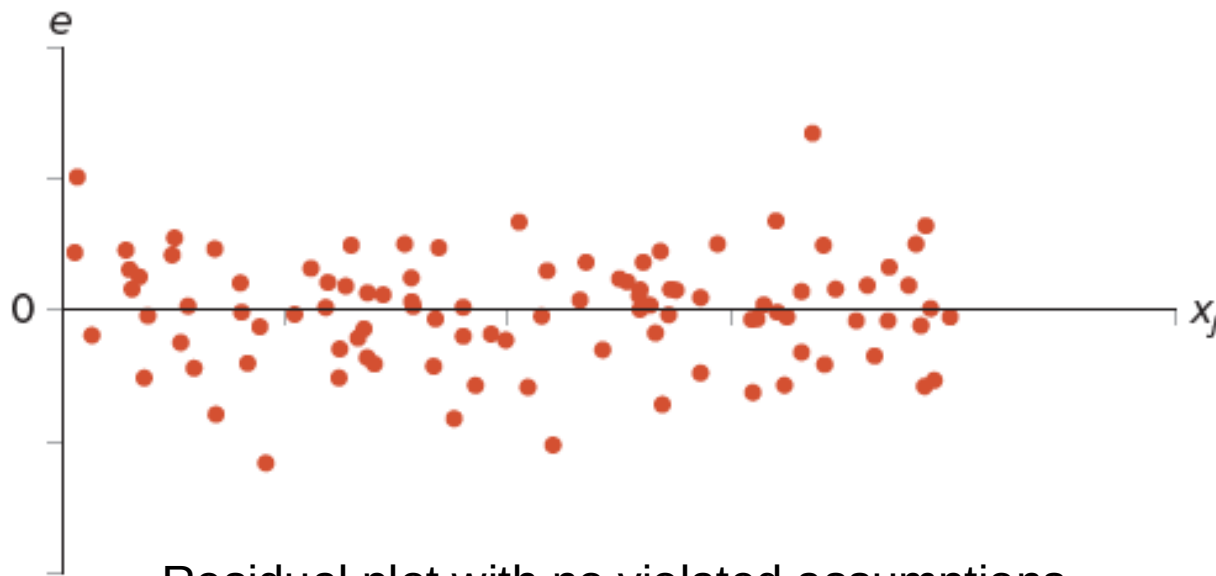


No linear relationship



Model Assumptions and Common Violations

Linearity is justified if the residuals are randomly dispersed.

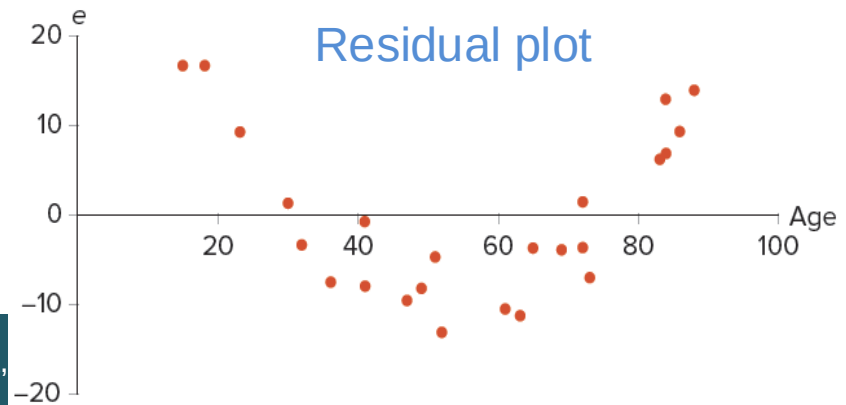
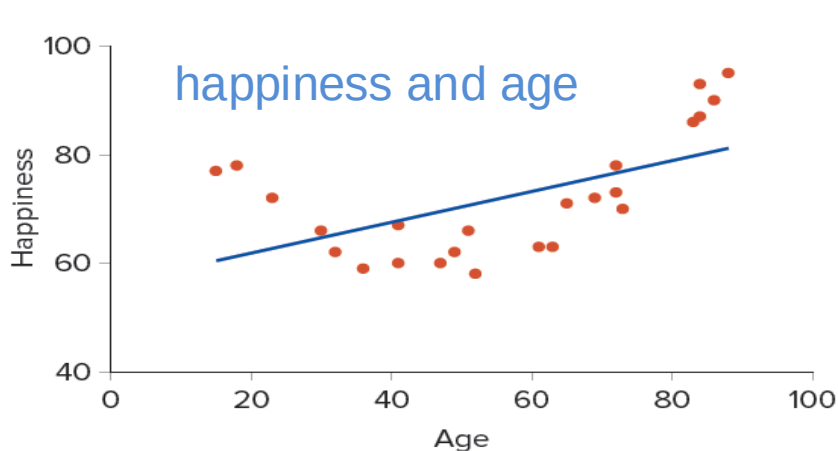


Residual plot with no violated assumptions

Sample Violations

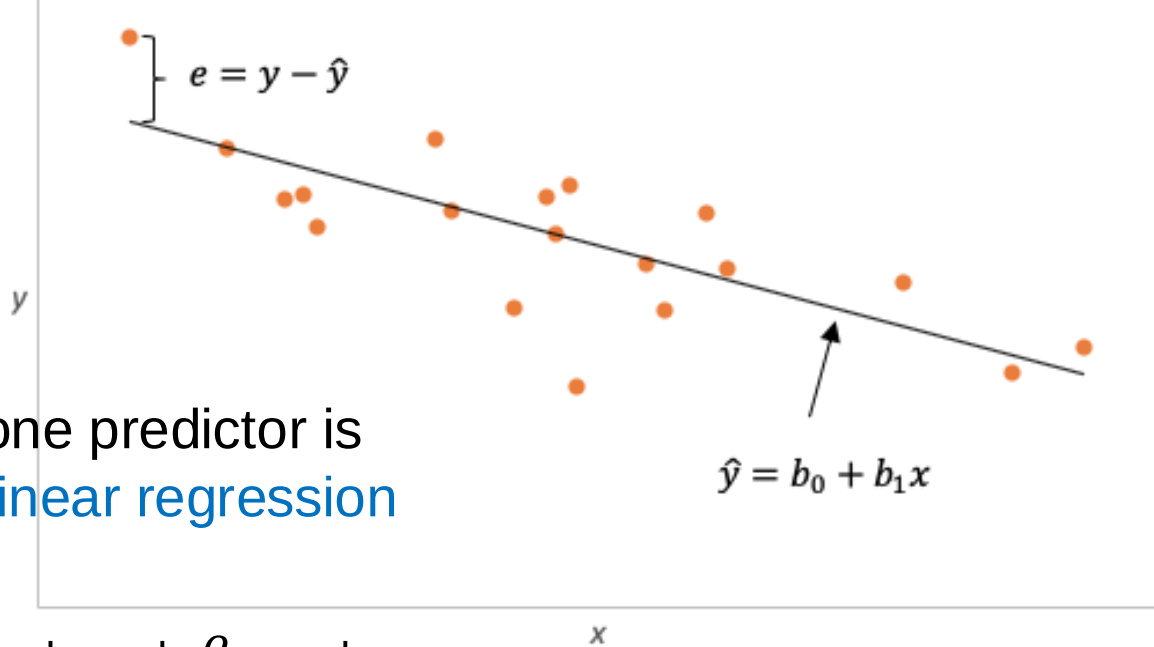
Residual with nonlinear Patterns

- Linear regression assumes a linear relationship between the response and predictors.
- If violated, OLS estimates and inferences can be misleading.
- **Detection**
 - Use residual plots to identify nonlinear patterns.
 - A discernible trend in the residuals indicates a nonlinear pattern.
- **Remedy**
 - Employ nonlinear regression.
 - Transform the response and predictor variables.



Multiple Linear Regression Model

- A model with more than one predictor is referred to as a **multiple linear regression** model.

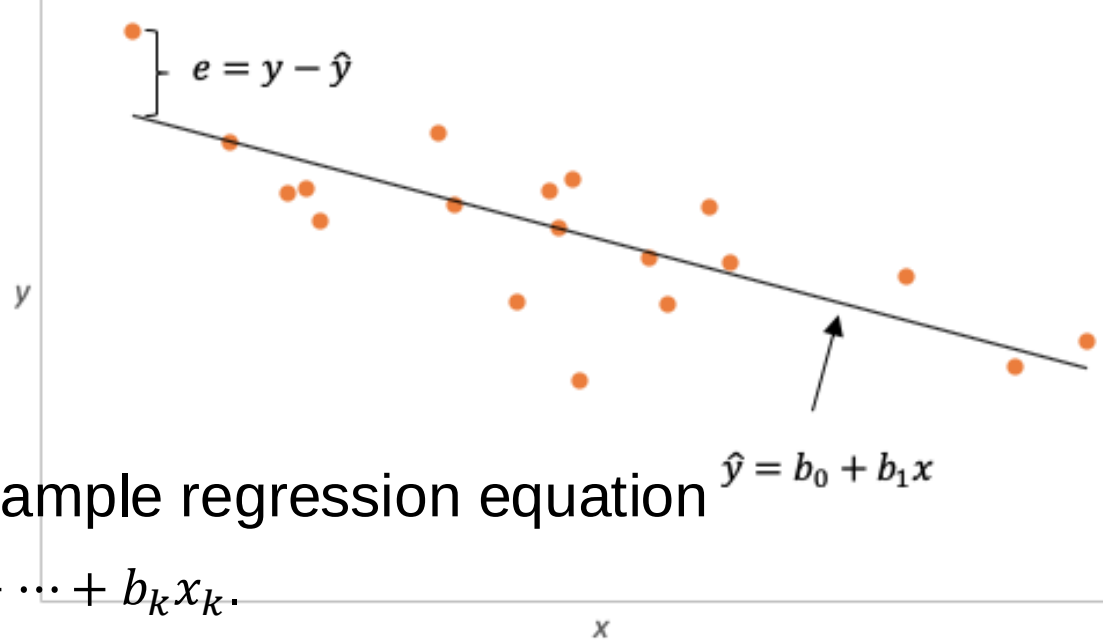


$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

- The population **parameters** $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are unknown and **must be estimated** using sample data.
- Use the sample data to obtain $b_0, b_1, b_2, \dots, b_k$ which are estimates of $\beta_0, \beta_1, \beta_2, \dots, \beta_k$.

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k.$$

The Linear Regression Model



- $b_0, b_1, b_2, \dots, b_k$ form the sample regression equation
$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots + b_kx_k.$$
- $\hat{y}$ (read as y-hat) is the **predicted value** of the **response variable** given specified values of the **predictor variables**.
- The difference between the observed and the predicted values is the **residual**: $e = y - \hat{y}$. → **error of prediction**
- **Ordinary Least Squares** (OLS) chooses the sample regression equation by minimizing the error sum of squares, $SSE = \sum (y - \hat{y})^2 = \sum e^2$.

The Linear Regression Model

- Interpret the estimated regression coefficients.
- b_j is the estimate of β_j
 - Change in the predicted value of the response, given a unit increase in x_j , holding all other predictor variables constant
 - *Partial influence* of x_j on $\hat{y}$
- b_0 is the estimate of β_0
 - Predicted value of $\hat{y}$ when each predictor variable assumes a value of 0
 - Not always meaningful

The Linear Regression Model: Excel function

- Example: $Earnings = \beta_0 + \beta_1 Cost + \beta_2 Grad + \beta_3 Debt + \beta_4 City + \varepsilon$

School ^a	Earnings	Cost	Grad	Debt	City
St. Ambrose Univ.	44800	22920	62	88	1
Albion College	45100	23429	73	92	0
⋮	⋮	⋮	⋮	⋮	⋮
Wittenburg Univ.	42700	26616	64	90	1

Notes: All data were downloaded from <https://collegescorecard.ed.gov> from May 24–26, 2016; see website for complete description of data.

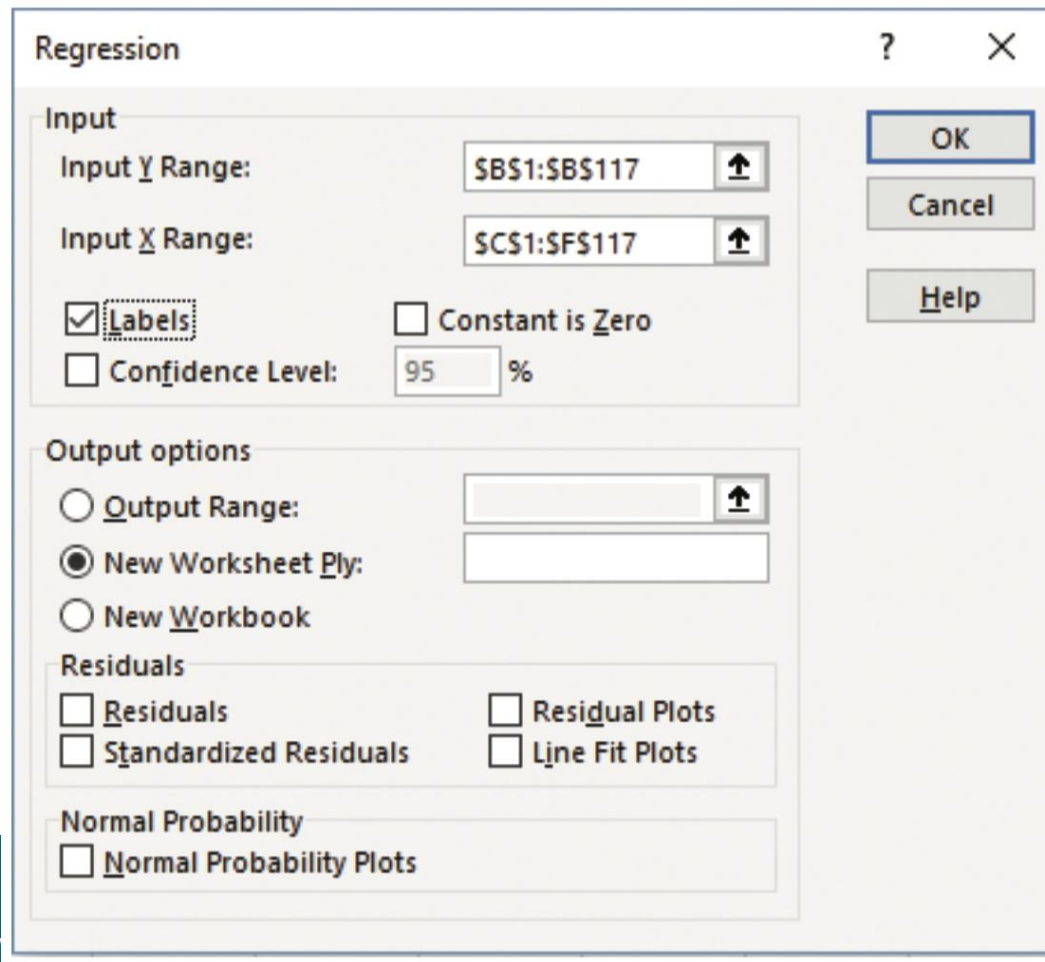
- What is the sample regression equation?
- Interpret b_j (the slope coefficients).
- Predict the annual post-college earnings if a college's average annual cost is \$25,000, its graduation rate is 60%, its percentage of students paying down debt is 80%, and it is located in a city.

Treatment of categorical data

- A regression model **treats all predictor variables as numerical** (assumes meaningful values)
- A *categorical variable* may need to be transformed into, e.g. ordered number, bins or one-hot variables (independent values)
- For one-hot or dummy variables denoted d .
 - For binary-value variable
 - $d = 1$ for one of the categories
 - $d = 0$ for the other(s)
 - Example: City is 1 if a city location, 0 otherwise
 - Can be multi-values.

7.1: The Linear Regression Model (15/26)

- Example continued with Excel
- Open the College data file
- Data > Data Analysis > Regression from the menu



The image shows the 'Regression' dialog box in Microsoft Excel. The dialog box is titled 'Regression' and has a question mark icon and a close button (X) in the top right corner. It is divided into several sections:

- Input:**
 - Input Y Range:** SBS1:SBS117
 - Input X Range:** SCS1:SFS117
 - ☒ **Labels**
 - ☐ **Confidence Level:** 95 %
 - ☐ **Constant is Zero**
- Output options:**
 - ☐ **Output Range:** (empty box)
 - ☒ **New Worksheet Ply:** (empty box)
 - ☐ **New Workbook**
- Residuals:**
 - ☐ **Residuals**
 - ☐ **Standardized Residuals**
 - ☐ **Residual Plots**
 - ☐ **Line Fit Plots**
- Normal Probability:**
 - ☐ **Normal Probability Plots**

On the right side of the dialog box, there are three buttons: **OK**, **Cancel**, and **Help**.

The Linear Regression Model

Regression Statistics						
Multiple R	0.6552					
R Square	0.4292					
Adjusted R Square	0.4087					
Standard Error	5645.831					
Observations	116					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	4	2660691959	665172990	20.868	7.56E-13	
Residual	111	3538169765	31875403			
Total	115	6198861724				
	Coefficients	Standard Error	t Stat	p-value	Lower 95%	Upper 95%
Intercept	10004.9665	7634.3338	1.311	0.1927	-5122.98	25132.91
Cost	0.4349	0.1110	3.917	0.0002	0.21	0.65
Grad	178.0989	69.1940	2.574	0.0114	40.99	315.21
Debt	141.4783	117.2120	1.207	0.2300	-90.79	373.74
City	2526.7888	1103.4026	2.290	0.0239	340.32	4713.25

The Linear Regression Model for the case of the Graduate of Colege

- a. $\widehat{Earnings} = 10004.97 + 0.4349Cost + 178.10Grad + 141.48Debt + 2526.79City$
- b. All coefficients are *positive*, suggesting a *positive influence* of each predictor variable on the response variable.
- Holding all other predictor variables constant, if the average annual costs increase by \$1, then, on average, predicted earnings are expected to increase by \$0.4349.
 - So, all else constant, predicted earnings are \$2,526.79 for a higher graduate who lives in a city.
- c. $\widehat{Earnings} = 10,004.97 + 0.434 * 25,000 + 178.10 * 60 + 141.48 * 80 + 2,526.79 * 1 = 45,408.7991$ or about \$45,409

Model Selection

For the introductory case consider three models. Which should we choose?

Model 1: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \varepsilon$

Model 2: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \beta_2 \text{Grad} + \beta_3 \text{Debt} + \varepsilon$

Model 3: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \beta_2 \text{Grad} + \beta_3 \text{Debt} + \beta_4 \text{City} + \varepsilon$

- Several “goodness-of-fit” measures summarize how well the sample regression equation fits the data.
 - The standard error of the estimate, s_e
 - The coefficient of determination, R^2
 - The adjusted coefficient of determination, adjusted R^2

Model Selection

- A residual or error = difference between the observed and predicted value of the response, $e_i = y_i - \hat{y}_i$.
- The standard error, s_e , is derived from the average squared deviation between the observed and predicted values.

$$s_e = \sqrt{\frac{SSE}{n - k - 1}}$$

- SSE (Sum Square of Error) = $\sum e_i^2$
- $n - k - 1$ is the degrees of freedom for the s_e where:
 - n is the sample size.
 - k denotes the number of predictors (x_i).
- When comparing models with the same response (Y), the model with the smaller s_e (less dispersion of the residual) is preferred.

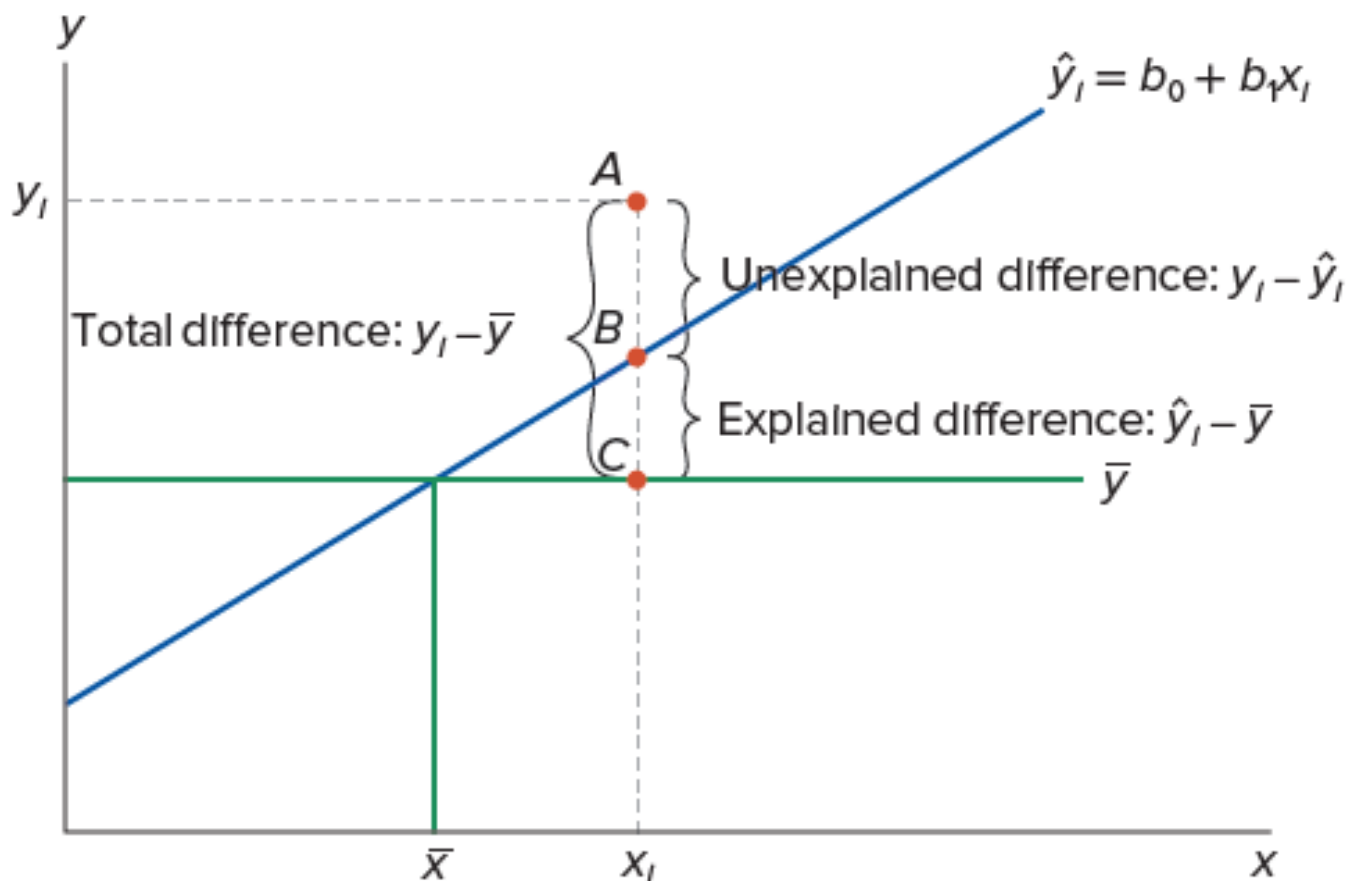
Model Selection

- The **coefficient of determination**, R^2 , quantifies the sample variation in the response.
- Use analysis of variance (**ANOVA**) in the context of regression:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- Usually, falls between 0 and 1
- The closer to 1, the better the fit
- $SST = \sum (y_i - \bar{y})^2$: The Sum Square of Total variation in y ,
- $SSR = \sum (\hat{y}_i - \bar{y})^2$: The Sum Square of variation explained by the Regression
- $SSE = \sum (y_i - \hat{y}_i)^2$: The Sum Square of Prediction Error
- $SST = SSR + SSE$
 - explain variation of system by the regression and unexplained variation

6.2: Model Selection (4/7)



$$\begin{aligned} A &= Y_i \\ B &= \hat{y} \\ C &= \bar{y} \end{aligned}$$

Model Selection When models have different number of variables

- We cannot use R^2 for model comparison when the competing models do not have the same number of predictor variables
 - R^2 **never decreases** as we add more variables.
 - Adding more variables with no economic or intuitive foundation is no good.
- **Adjusted R^2** explicitly accounts for the sample size n and the number of predictor variables k .

$$= 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1} \right)$$

- Imposes a **penalty for any additional predictors**.
- The higher the **adjusted R^2** , the better the model.

Model Selection

- Example: Recall the introductory case and consider three models.

Model 1: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \varepsilon$

Model 2: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \beta_2 \text{Grad} + \beta_3 \text{Debt} + \varepsilon$

Model 3: $\text{Earnings} = \beta_0 + \beta_1 \text{Cost} + \beta_2 \text{Grad} + \beta_3 \text{Debt} + \beta_4 \text{City} + \varepsilon$

	Model 1	Model 2	Model 3
Standard error of the estimate s_e	6271.4407	5751.8065	5645.8306
Coefficient of determination R^2	0.2767	0.4023	0.4292
Adjusted R^2	0.2703	0.3862	0.4087

- Which of the three models is the preferred model?
- Interpret the coefficient of determination for the preferred model.
- What percentage of the sample variation in annual post-college earnings is unexplained by the preferred model?

Model Selection

	Model 1	Model 2	Model 3
Standard error of the estimate s_e	6271.4407	5751.8065	5645.8306
Coefficient of determination R^2	0.2767	0.4023	0.4292
Adjusted R^2	0.2703	0.3862	0.4087

- a. Model 3, since it has the lowest standard error of the estimate and the **highest adjusted R^2** .
- b. Model 3 explains 42.92% of the sample variation in the earnings. → using **R^2**
- c. Model 3 does not explain 57.08% of the sample variation in earnings. → using **R^2**

Tests of Significance

- Consider the linear regression model with k predictor variables: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$.
- We can conduct hypothesis tests about the unknown parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_k$.
 - Joint test about all of the parameters
 - Individual tests about a single parameter
- For the tests to be valid, $b_0, b_1, b_2, \dots, b_k$ must be normally distributed.
 - Satisfied if the random error term ε is normally distributed
 - If ε is not normally distributed, the tests can be assumed valid only for large sample sizes.

Tests of Significance: Joint Test

- A test of joint significance → a test of the overall usefulness of a regression.
 - whether the predictor variables have a joint statistical influence on the response.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

Hypothesis

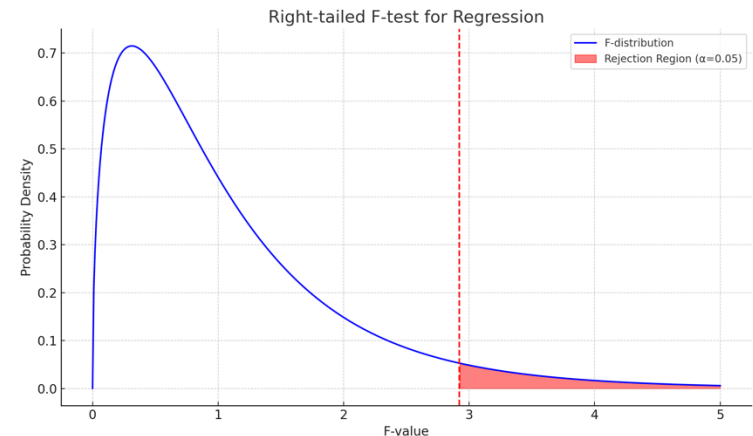
$$H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_A: \text{At least one } \beta_j \neq 0$$

- H_0 : All of the slope coefficients equal zero: all of the predictors drop out; none of the predictors have a linear relationship with the response.
- H_A : At least one of the slope coefficients does not equal zero: at least one predictor influences the response.

Tests of Significance

- The test statistic is $F_{(df_1, df_2)} = \frac{SSR/k}{SSE/(n-k-1)} = \frac{MSR}{MSE}$
 - $df_1 = k$
 - $df_2 = n - k - 1$
 - SSR is the regression sum of squares.
 - SSE is the error sum of squares.
 - MSR is the mean square regression.
 - MSE is the mean square error.



- This is a right-tailed F test; the **p-value** is $P\left(F_{(df_1, df_2)} \geq \frac{MSR}{MSE}\right)$.
 - A larger test statistic provides us with more evidence to reject the null hypothesis.
 - A larger test statistic indicates that a large portion of the sample variation in the response is explained by the regression model, thus the model is useful.

Tests of Significance

ANOVA	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	Significance <i>F</i>
Regression	<i>k</i>	<i>SSR</i>	$MSR = \frac{SSR}{k}$	$F_{(df_1, df_2)} = \frac{MSR}{MSE}$	$P\left(F_{(df_1, df_2)} \geq \frac{MSR}{MSE}\right)$
Residual	$n - k - 1$	<i>SSE</i>	$MSE = \frac{SSE}{n - k - 1}$		
Total	$n - 1$	<i>SST</i>			

Tests of Significance

- Example: Model 3 was the preferred model.
- Conduct a hypothesis test to determine if the predictor variables are jointly significant in explaining earnings.

ANOVA	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	Significance <i>F</i>
Regression	4	2660691959	665172990	20.868	7.56E-13
Residual	111	3538169765	31875403		
Total	115	6198861724			

For Excel, 1-F.DIST(20.868,4,111,TRUE)
Or F.DIST.RT()

(F.distribution, Right tail)

Tests of Significance

- We set up the following competing hypotheses:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$H_A: \text{At least one } \beta_j \neq 0$$

- $n = 116, k = 4, df_1 = 4, df_2 = 111$, and $\frac{MSR}{MSE} = 20.868$
- The p-value, $P(F_{(4,111)} \geq 20.868)$, is approximately zero (< 0.05).
- So, **we Reject H_0** . At the 5% significance level, the predictor variables are jointly significant in explaining Earnings.

Tests of Significance: Individual test

- Consider the linear regression model with k predictor variables.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

- If $\beta_1 = 0$, then x_1 basically drops out of the equation.
 - There is no linear relationship between x_1 and y .
 - x_1 does not influence y .
- Let β_{j0} be a hypothesized value of β_j . The typical hypothesis for a test of individual significance is:

Two-Tailed Test

$$H_0: \beta_j = \beta_{j0}$$

$$H_A: \beta_j \neq \beta_{j0}$$

Tests of Significance (use a t-Test)

- The test statistic for a test of individual significance is $t_{df} = \frac{b_j - \beta_{j0}}{se(b_j)}$.
 - $df = n - k - 1$
 - b_j is the estimate for β_j .
 - $se(b_j)$ is the standard error of the estimator b_j .
 - β_{j0} is the hypothesized value of β_j .
 - If the hypothesized value is zero, $t_{df} = \frac{b_j}{se(b_j)}$.
- Note that the testing framework for the intercept β_0 is similar.
- *Statistical packages report test statistics and p-values for a two-tailed test that assesses whether the coefficient differs from zero.*

Two-Tailed Test

$$H_0: \beta_j = \beta_{j0}$$

$$H_A: \beta_j \neq \beta_{j0}$$

Tests of Significance

- Example: Model 3 was the preferred model.
- Conduct a hypothesis test to determine whether cost influences earnings at the 5% significance level.

	Coefficients	Standard Error	t Stat	p-value
Intercept	10004.9665	7634.3338	1.311	0.1927
Cost	0.4349	0.1110	3.917	0.0002
Grad	178.0989	69.1940	2.574	0.0114
Debt	141.4783	117.2120	1.207	0.2300
City	2526.7888	1103.4026	2.290	0.0239

Tests of Significance

	Coefficients	Standard Error	t Stat	p-value
Intercept	10004.9665	7634.3338	1.311	0.1927
Cost	0.4349	0.1110	3.917	0.0002
Grad	178.0989	69.1940	2.574	0.0114
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- We set up the following competing hypotheses to determine whether costs influence earnings (test for b_1).
$$H_0: \beta_1 = 0$$
$$H_A: \beta_1 \neq 0$$
- $b_1 = 0.4349$, $s_e(b_1) = 0.1110$, $t = \frac{0.4349-0}{0.1110} = 3.917$
- The p-value is 0.0002 (< 0.05), $df = 111$, so we reject H_0 .
- At the 5% significance level, Cost is significant in explaining Earnings.
- For Excel, $T.DIST.2T(3.917,111) = 0.000155$

Model Selection

- Regression results are often reported in a table.

			Model 1			Model 2			Model 3
Intercept			28,375.4051* (0.000)			11819.4747 (0.129)			10,004.9665 (0.193)
Cost			0.7169* (0.000)			0.5050* (0.000)			0.4349* (0.000)
Grad			NA			192.6664* (0.007)			178.0989* (0.011)
Debt			NA			104.6573 (0.378)			141.4783 (0.230)
City			NA			NA			2,526.7888* (0.024)
Se			6,271.4407			5,751.8065			5,645.8306
R^2			0.2767			0.4023			0.4292
Adjusted R^2			0.2703			0.3862			0.4087
F-test (p-value)			43.608 (0.000)			25.124 (0.000)			20.868(0.000)

NOTES: Parameter estimates are in the top half of the table with the p -values in parentheses; * represents significance at the 5% level. NA denotes not applicable. The lower part of the table contains goodness-of-fit measures.

Trials with only Significant

Variables

SUMMARY OUTPUT (No Intercept and Debt)

Regression Statistics					
Multiple R	0.991083				
R Square	0.982246				
Adjusted R Square	0.973082				
Standard Error	6351.743				
Observations	116				
ANOVA					
	df	SS	MS	F	Significance F
Regression	3	2.52219E+11	84073058573	2083.872	4.7E-99
Residual	113	4558944282	40344639.67		
Total	116	2.56778E+11			
	Coefficients				
	s	Standard Error	t Stat	P-value	Lower 95%
Intercept	0	#N/A	#N/A	#N/A	#N/A
Cost	0.711044	0.112154513	6.339864965	4.87E-09	0.4
Grad	411.2786	41.83479304	9.831018763	7.46E-17	328
City	2065.424	1228.320116	1.681502736	0.095427	

SUMMARY OUTPUT (On Intercept, Debt and City)

Regression Statistics					
Multiple R	0.990859				
R Square	0.981801				
Adjusted R Square	0.97287				
Standard Error	6402.451				
Observations	116				
ANOVA					
	df	SS	MS	F	Significance F
Regression	2	2.52105E+11	1.26053E+11	3075.099	3E-99
Residual	114	467301679	40991375.37		
Total	116	2.56778E+11			
	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	0	#N/A	#N/A	#N/A	#N/A
Cost	0.761975	0.1088493	7.000271334	1.88E-10	0.546345
Grad	407.8026	42.117257	9.682553225	1.53E-16	324.3686

Conclusions

LO 1 The linear regression model is introduced for predicting numeric outcome and also the interpretation of the result.

LO 2 Select a good model by interpreting the goodness-of-fit measures, esp. R^2 and adjusted R^2 .

LO 3 Conduct tests of significance, e.g. p-value, F-stats to validate the importance of predictors.